

PROJECT 1 — PARAMETRIC MOUNTING BRACKET

Project Type: CAD / 3D Modelling Capstone

Client Brief

Design a small aluminium equipment mounting bracket suitable for CNC milling. The component must be driven by a clear set of parameters so that mounting-hole spacing and major envelope dimensions can be changed without rebuilding the model.

Design Specification

- Overall envelope: approximately 120 × 60 × 80 mm
- Base and vertical plate thickness: 8 mm
- Four base mounting holes
- Two component mounting holes on the vertical face
- Two triangular/gusset reinforcements
- External fillets and selected chamfers
- Material: Aluminium

Parametric Requirements

BRACKET_LENGTH, BRACKET_WIDTH, BRACKET_HEIGHT, PLATE_THICKNESS, MOUNT_HOLE_DIA, MOUNT_HOLE_SPACING_X, MOUNT_HOLE_SPACING_Y, COMPONENT_HOLE_DIA, COMPONENT_HOLE_SPACING, GUSSET_THICKNESS, EDGE_FILLET.

Required Skills

- Fully constrained sketches
- Geometric and dimensional constraints
- Pad/Extrude, Pocket/Cut, Fillet, Chamfer, Mirror, Pattern
- Reference/datum geometry
- Feature-tree organization
- Parametric expressions and design intent
- 2D drawing → 3D model → 2D drawing

Client Approval Questions

- What is the design intent?
- Which dimensions are driving parameters?
- Is every functional sketch fully constrained?
- What happens when hole spacing or plate thickness changes?
- How is the final solid validated?
- Does the drawing accurately represent the current model?

CAD QA / Validation

- Solid validity / closed geometry

- Sketch constraint status
- Parametric regeneration test
- Feature dependency review
- Drawing/model consistency
- STEP export and re-import
- STL manifold/watertight check

Required Deliverables

- Native FreeCAD model (.FCStd)
- Manufacturing / engineering drawing (PDF)
- STEP export
- STL export where relevant
- Parameter/design-intent record
- CAD QA checklist/report